BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 4483 / 4484

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

CHEMISTRY PAPER-II GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 4483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. Formula of chloroform is:

(A) CH_3Cl (B) CCl_4

(C) CH_2Cl_2 (D) $CHCl_3$

Q2. Benzene cannot undergo reaction like:

(A) Elimination (B) Addition

(C) Oxidation (D) Substitution

Q3. Which of given is electrophile?

(A) NH_3 (B) H_2O

(C) BF_3 (D) Cl_2

Q4. Which compound shows strong hydrogen bonding?

(A) C_2H_6 (B) C_2H_4

(C) $C_2H_5-O-C_2H_5$ (D) C_2H_5OH

Q5. Which of the given compound will react with Tollen's reagent?

(A) $CH_3-C(=O)-OH$ (B) $CH_3-C(=O)-H$

(C) $CH_3-C(=O)-CH_3$ (D) CH_3-O-CH_3

Q6. Which of given is not fatty acid?

(A) Propanoic acid
(B) Acetic acid
(C) Phthalic acid
(D) Butanoic acid

Q7. Which of these polymers is synthetic polymer?

(A) Polyester
(B) Starch
(C) Animal fat
(D) Cellulose

Q8. Temperature of decomposition zone during manufacturing of cement goes upto:

(A) 600°C
(B) 800°C
(C) 1000°C
(D) 1200°C

Q9. To avoid the formation of toxic compounds with chlorine which substance is used for disinfecting water?

(A) $KMnO_4$
(B) Chloramines
(C) Alums
(D) O_3

Q10. In water the concentration of dissolved O_2 should be:

(A) 1 – 3 ppm
(B) 2 – 4 ppm
(C) 4 – 8 ppm
(D) 8 – 12 ppm

Q11. Which statement is correct?

(A) Na atom is smaller than Na^+
(B) Na atom is larger than K atom
(C) F atom is smaller than F^-
(D) F atom is larger than F^-

Q12. Chile saltpetre has the chemical formula:

(A) $NaNO_3$
(B) KNO_2
(C) $Na_2B_4O_7$
(D) $Na_2CO_3.H_2O$

Q13. Which element belongs to group IV-A of the Periodic table?

(A) Barium
(B) Sodium
(C) Lead
(D) Oxygen

Q14. Elements of group VI-A also called:

(A) Halogens
(B) Chalogens
(C) Chalite
(D) Halite

Q15. An element having high ionization energy and tends to be chemically inactive is:

(A) An alkali metal
(B) Halogen
(C) Noble gas
(D) Transition element

Q16. Which is the correct formula of Tetraammine Chloro-nitro-Platinum(IV) sulphate?

(A) $[PtCl(NO_2)(NH_3)_4]SO_4$ (B) $[Pt(NO_2)Cl(NH_3)_4]SO_4$

(C) $[Pt(NH_3)_4Cl(NO_2)]SO_4$ (D) $[Pt(NH_3)_4(NO_2)Cl]SO_4$

Q17. A double bond consists of:

(A) Two sigma bonds (B) Two Pi bonds

(C) One sigma and one Pi bond (D) One sigma and two Pi bonds

CHEMISTRY PAPER-II GROUP-I (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) Define ionization energy with an example.

(ii) Write down any two dissimilarities of Hydrogen with group 1-A elements.

(iii) How is gypsum converted into Plaster of Paris?

(iv) Write down the formulas of (i) Dolomite (ii) Halite

(v) How chromate ions are converted into dichromate ions?

(vi) Why does damaged tin plated iron get rusted quickly?

(vii) Elaborate the mechanism of S_n 2 reactions.

(viii) Define nucleophile with an example

(ix) Draw the structure of cholesterol.

(x) How vinyl acetate converted into polyvinyl acetate.

(xi) Write down the name of any four classes of enzymes.

(xii) Mention the role of Phosphorus in early growth of plant.

Q3. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) How alkene is converted into epoxide? What is its application?

(ii) Prepare the cyclic polymer of ethyne.

(iii) How good quality polythene is obtained from ethene?

(iv) How does H_3PO_3 act as a reducing agents?

(v) Give four uses of H_2SO_4

(vi) Write the names and examples of two compounds containing carbonyl functional group.

(vii) Name two types of the isomerism shown by alkene with example.

(viii) Justify that bleaching powder is oxidizing agent.

(ix) What are Freon and Teflon?

(x) What chemical reaction takes place in stratosphere with ozone?

(xi) What is Smog? Give its types?

(xii) What are leachates?

Q4. (Marks 12) Attempt any six parts. 6 × 2 =

(i) [2] Why CO_2 is gas while SiO_2 is solid?

(ii) [1+1=2] Write down any two uses of Al.

(iii) [2] What is the chemistry of borax bead test?

(iv) [1+0.5+0.5=2] Define resonance. Write down Kekule's structures of benzene.

(v) [2] What is denaturing of alcohol?

(vi) [2] Why is Phenol acidic in nature?

(vii) [1+0.5+0.5=2] What is formalin? Give its two uses.

(viii) [1+1=2] Write down the structures of: (a) Malonic acid (b) Phthalic acid

(ix) [2] What is strecker synthesis?

Q. SECTION-II

QNOTE. (Marks 3 × 8 = 24) Attempt any three questions.

Q5.

(a) [4] Write down the point of similarities and difference of hydrogen with IA and IVA groups. (any two of each)

(b) [4] Describe with diagram the manufacture of sodium by Down's Cell.

Q6.

(a) [4] Give any eight applications of Noble gases.

(b) [4] How do Diammonium phosphate and Potassium nitrate prepared? Give their properties and uses.

Q7.

(a) [1+3=4] What is Cracking of petroleum? Discuss its three types.

(b) [4] Explain two main factors which govern the reactivity of alkyl halides.

Q8.

(a) [4] Describe both Linear Polymerization and Cyclic polymerization of Acetylene by means of chemical reaction.

(b) [4] Write a note on Aldol condensation reaction of carbonyl compounds with mechanism.

Q9.

(a) [4] Describe structure of benzene on the basis of atomic orbital treatment.

(b) [4] How ethyl alcohol is prepared by the fermentation of: (i) Molasses (ii) Starch

GROUP – II

CHEMISTRY PAPER-II GROUP-II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 4484

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. The fibre which is made from acrylonitrile as monomer:

(A) PVC
(B) Rayon fibre
(C) Acrylic fibre
(D) Polyester fibre

Q2. For which crop ammonium nitrate fertilizer is not used?

(A) Cotton
(B) Wheat
(C) Sugar cane
(D) Paddy rice

Q3. Which of following is better to disinfect water?

(A) Cl_2
(B) O_2
(C) O_3
(D) $KMnO_4$

Q4. The main pollutant of leather tanneries in the waste water is due to the salt of:

(A) Lead
(B) Chromium(VI)
(C) Copper
(D) Chromium(III)

Q5. Zn, Cd, Hg in Mendeleev's periodic table, were placed with:

(A) Noble metals
(B) Alkaline earth metals
(C) Inner transition metals
(D) Coinage metals

Q6. Down's cell is used to prepare:

(A) Sodium carbonate
(B) Sodium hydroxide
(C) Sodium bicarbonate
(D) Sodium metal

Q7. Boric acid cannot be used:

(A) As antiseptic in medicine
(B) For washing eyes
(C) In soda bottles
(D) For enamels and glazes

Q8. An element that has a high ionization energy and tends to be chemically inactive would most likely to be:

(A) A noble gas
(B) A transition element
(C) An alkali metal
(D) A halogen

Q9. Formic acid on reaction with dehydrating agent give:

(A) CO_2,CO,H_2O
(B) CO, OH^-
(C) CO, H_2O
(D) CO and CO_2

Q10. The strength of binding energy of transition elements depends upon:

(A) Number of electron pairs
(B) Number of unpaired electrons
(C) Number of neutrons
(D) Number of protons

Q11. The state of hybridization of carbon atom in alkane is:

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q12. $H_2C{=}CH{-}C{\equiv}CH$ and conc HCl on reaction give:

(A) Polyacetylene
(B) Benzene
(C) Chloroprene
(D) Divinyl acetylene

Q13. Amongst the following, the compound that can be most readily sulphonated is:

(A) Toluene
(B) Benzene
(C) Nitrobenzene
(D) Chlorobenzene

Q14. Which one is more reactive alkyl halide?

(A) R−F
(B) R−Cl
(C) R−Br
(D) R−I

Q15. Methyl alcohol is not used:

(A) As a solvent
(B) As a substitute for petrol
(C) As an anti-freezing agent
(D) For denaturing of ethyl alcohol

Q16. Acetone reacts with HCN to form a cyanohydrin, it is an example of:

(A) Electrophilic addition
(B) Electrophilic substitution
(C) Nucleophilic addition
(D) Nucleophilic substitution

Q17. Which acid is used in the manufacturing of synthetic fibre?

(A) Formic acid
(B) Oxalic acid
(C) Carbonic acid
(D) Acetic acid

CHEMISTRY PAPER-II GROUP-II (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) Why are the ionic radii of negative ions larger than the size of their parent atoms? Give example.

(ii) Why does the oxidation state of noble gases usually zero?

(iii) Give reactions of lithium with oxygen and carbon dioxide.

(iv) What are the products formed when magnesium reacts with nitrogen and sulphur?

(v) Why does damaged tin plated iron get rusted quickly?

(vi) How does the process of galvanizing protect from rusting

(vii) Give reaction of ethyl magnesium bromide with formaldehyde followed by acid hydrolysis

(viii) Give reaction for the preparation of ethyl alcohol from ethyl bromide. Also mention reaction conditions.

(ix) Define lipids. Give difference between fats and oils.

(x) Briefly describe the term "Specificity" of enzyme.

(xi) What is the difference between simple lipids and compound lipids?

(xii) Why are nitrogenous fertilizers supplied to the plants of soil?

Q3. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) NO_2 is strong oxidizing agent. Prove the truth of this statement giving two examples.

(ii) Complete and balance the given equations: (i) P + NO → ? (ii) $HNO_2 + CO(NH_3)_2$ → ?

(iii) Why HF is weaker acid than HCl?

(iv) How does iodine pentoxide (I_2O_5) react with H_2O and CO?

(v) What is Catenation?

(vi) What is Catalytic cracking?

(vii) Write down structural formulas of 1, 3 – Butadiene and 2 – methyl – 2 – butene.

(viii) Differentiate between clemmensen and wolf-Kishner reduction giving chemical reactions.

(ix) How would you prepare trans alkene from alkyne?

(x) What is ecosystem?

(xi) How do oxides of sulphur adversely affect the environment?

(xii) How is value of COD determined?

Q4. (Marks 12) Attempt any six parts. 6 × 2 =

(i) What are Silicones? How are they prepared?

(ii) What is Borax? Give its commercial preparation.

(iii) What is importance of oxides of Lead in Paints?

(iv) What were objections to Kekule's formula for Benzene?

(v) Ethyl alcohol is a liquid while methyl chloride is a gas. Why?

(vi) Water has higher boiling point than Ethanol. Justify.

(vii) How will you distinguish between Methanal and Ethanal?

(viii) What happens when Sodium formate is heated with Soda lime?

(ix) What are Essential and Non-essential Amino Acids?

Q. SECTION-II

QNOTE. (Marks 3 × 8 = 24) Attempt any three questions.

Q5.

(a) [4] Discuss the improvements made in the Mendeleev's Periodic Table and also discuss defects in the Mendeleev's Periodic Table.

(b) [4] Explain commercial preparation of Sodium metal by Down's cell and also give advantages of Down's cell.

Q6.

(a) [4] How bleaching powder is prepared by Hasenclever's method?

(b) [1+3=4] What is paper? Describe the process of digestion in paper industry.

Q7.

(a) [1+1+1+1=4] Define with example (i) Tautomerism (ii) Metamerism (iii) Position isomerism (iv) Functional group isomerism

(b) [1+3=4] What do you understand by the term Nucleophilic substitution? Explain S_n 2 mechanism in detail.

Q8.

(a) [4] Define Markownikov's rule. Predict the structures of the alcohol obtained by the addition of the acid to the given compounds: (i) Propene (ii) 1–Butene (iii) 2–Butene

(b) [4] How does acetaldehyde react with (i) $NaHSO_3$ (ii) Conc. NaOH (iii) HCN (iv) NH_2OH

Q9.

(a) [1+1+2=4] Define aromatic nitration along with example and its mechanism.

(b) [2+2=4] How ethanol is prepared from molasses and starch?